Open Source LLMs

General Agents, Chapter 2/12 — Revision & Viva-Prep Cheatsheet (English + Roman Urdu)
Source: agentfactory.panaversity.org/docs/open-source-llms

EN: Every AI tool has 2 parts: Harness (does the work) and Brain (the model). The harness reaches the brain at an address -- that address can point to your laptop (Ollama), a server (vLLM), or the cloud (OpenRouter) and the harness never knows the difference.
Roman Urdu: Har AI tool ke 2 parts hain: Harness (kaam karta) aur Brain (model). Harness brain ko ek address pe reach karta hai -- woh address laptop (Ollama), server (vLLM), ya cloud (OpenRouter) point kar sakta, harness kabhi farq nahi janta.

1. Local Tier (Ollama) + the Two Walls

- Brain is just an address (phone-number analogy) -- a local model is a DIFFERENT brain, not a smaller copy.
- **2 walls:** Capability (needs a tool-use-trained model, no hardware fixes it) vs Throughput (needs GPU, no smarter model fixes it).
- Context window trap: default is often just 4,096 tokens (silently trimmed) -- set num_ctx to 64,000+.

2. Server Tier (vLLM) -- Many Users, One Machine

- Ollama stays flat under concurrency (it's a one-person-laptop design); vLLM is an 'industrial kitchen' -- continuous batching + PagedAttention let it scale with load.
- vLLM doesn't make one user faster -- it keeps the machine fast as load grows (only moves the throughput wall).

3. Cloud Tier (OpenRouter) + Mini LLM Cloud

- 'Open' gives 3 things: no lock-in, choice of landlord, a floor under the future. OpenRouter is a gateway (1 address/key/bill).
- Working rule: default to price-performance, escalate to frontier only on a real, observed failure.
- A full mini-cloud stack (vLLM+LiteLLM+Postgres+Open WebUI) adds identity/budgets/spending records a bare inference engine lacks.

Key Terms Cheatsheet

One line each — English + Roman Urdu.

Term	English	Roman Urdu
Harness vs Brain	The tool that acts vs the model it calls at an address.	Kaam karne wala tool vs address par call hone wala model.
Capability wall	Fixed by a stronger/tool-trained model, not more hardware.	Strong/tool-trained model se fix hoti, zyada hardware se nahi.
Throughput wall	Fixed by more GPU, not a smarter model.	Zyada GPU se fix hoti, smarter model se nahi.
vLLM	A multi-user inference server using continuous batching + PagedAttention.	Multi-user inference server, continuous batching + PagedAttention use karta.
OpenRouter	A gateway giving one address/key/bill across many frontier models.	Gateway jo ek address/key/bill se kai frontier models deta.

Quick Self-Check

Question	Answer
Why is a local model called a 'different brain', not a smaller copy?	It's a genuinely distinct model, not a scaled-down version of the cloud model behind the same product name.
What's the difference between the capability wall and the throughput wall?	Capability needs a better/tool-trained model; throughput needs more GPU -- neither fix solves the other's problem.
Why doesn't vLLM make a single request faster?	It's designed to keep the machine fast UNDER LOAD (many concurrent users), not to speed up one request.
What's the context-window trap to watch for locally?	Many setups silently default to just 4,096 tokens -- you must explicitly raise num_ctx.

Where This Leads / Sources

Leads to: Agentic Coding, Context Layer crash course. Full appendix on building a mini LLM cloud in 03-appendix.

Source: docs/open-source-llms/ (this repo), content originally via Zia Tutor AI connector.